UPSA
UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
Scholarship with Professionalism

BCPC 112: INTRODUCTION TO BUSINESS STATISTICS

LECTURE SEVEN & EIGHT

Probability Distribution (Normal)

Learning Objectives

In this lecture, you learn to:

- Explain the characteristics of probability distributions.
- Distinguish between discrete and continuous random variables.
- Explain the characteristics of the normal distribution.
- Compute probabilities using the standard normal distribution.
- Apply the normal distribution to solve business problems.

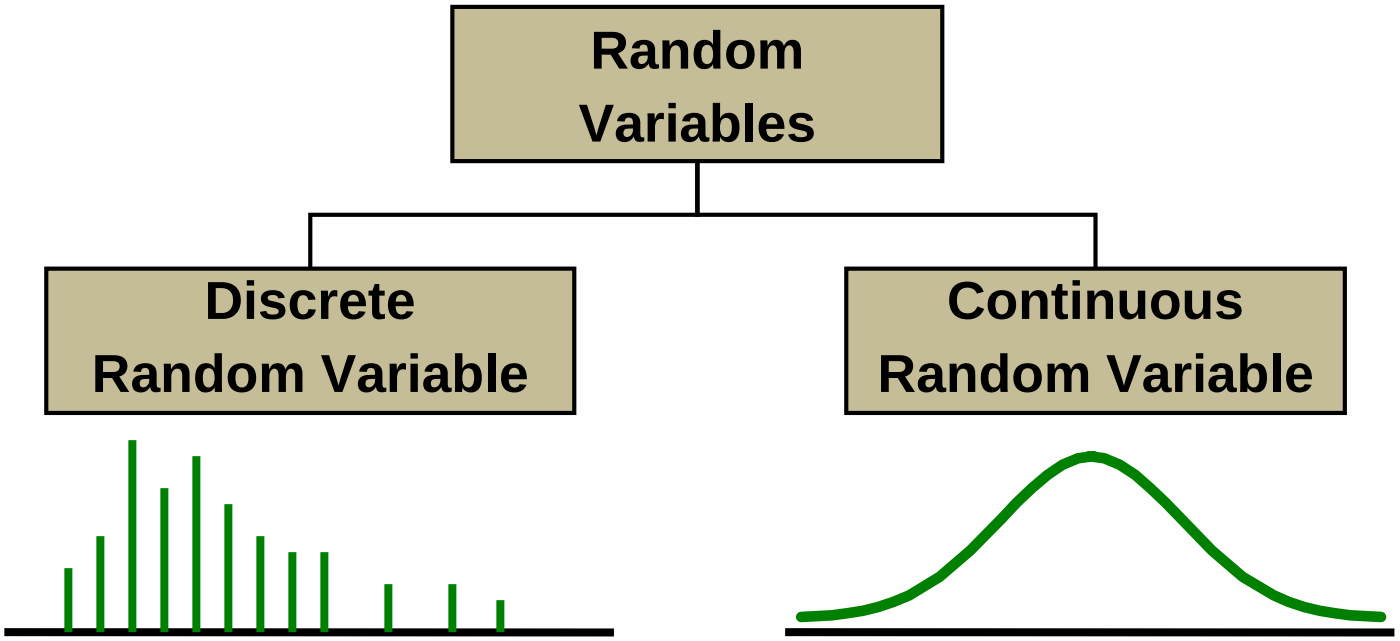
Recall

- What is probability?
- What is an event?
- What is a sample space?
- What is conditional probability?
- How did we calculate probabilities in Week 6?

Random Variables

- A **random variable** assigns a numerical value to every possible outcome of an experiment.
  - Usually denoted by upper-case letters, say X.
- **Discrete** random variables produce outcomes that come from a counting process.
  - Take integral values of X (e.g., number of customers waiting to be served).
- **Continuous** random variables produce outcomes that come from a measurement.
  - The continuous random variable X assumes many values within an interval of real numbers (e.g., the amount of sugar in a bottle of drink).

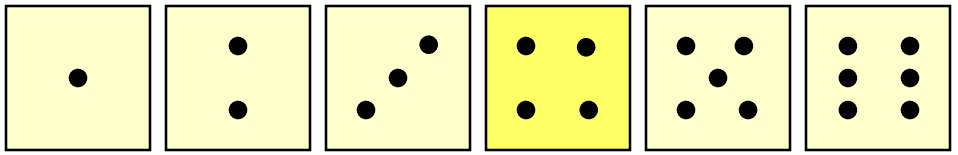
Random Variables



Discrete Random Variables

- Can only assume a countable number of values

Examples:



- **Roll a die twice**
  Let X be the number of times 4 occurs
  (then X could be 0, 1, or 2 times)

- **Toss a coin 4 times.**
  Let X be the number of heads
  (then X = 0, 1, 2, 3, or 4)



Discrete Random Variables

- Number of loan applications received today
- Number of customers entering a bank
- Number of defective products
- Number of insurance claims

Probability Distribution
(for a Discrete Random Variable)

- A **probability distribution for a discrete random variable** is a mutually exclusive listing of all possible numerical outcomes for that variable and a probability of occurrence associated with each outcome.

Number of courses taken	Probability
2	0.20
3	0.40
4	0.24
5	0.16

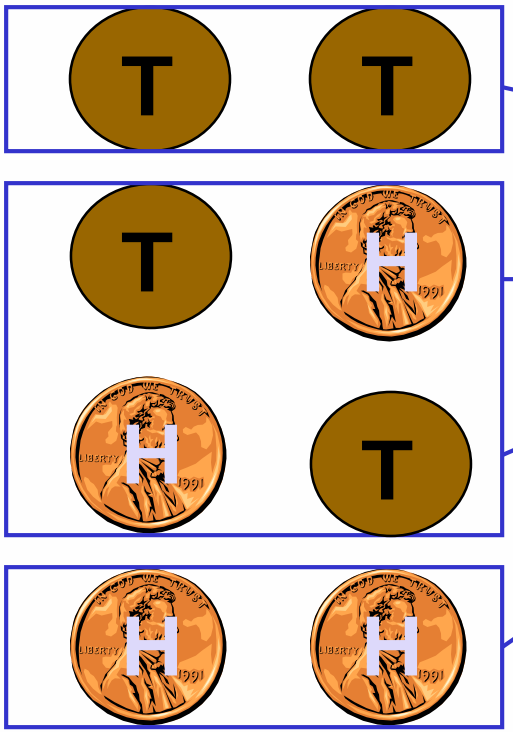
For any probability distribution:

1. $0 \leq P(X) \leq 1$
2. $\sum P(X) = 1$

Probability Distribution
(for a Discrete Random Variable)

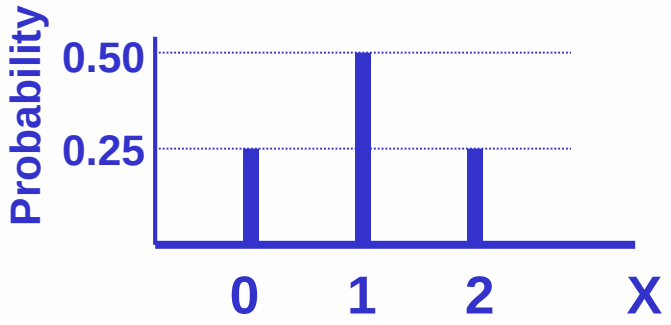
Experiment: Toss 2 Coins. Let X = # of heads.

4 possible outcomes



Probability Distribution

X Value	Probability
0	1/4 = 0.25
1	2/4 = 0.50
2	1/4 = 0.25



Probability Distribution
(for a Discrete Random Variable)

Example 7.1

The number of telephone calls received in an office between 12:00 noon and 1:00 pm has the probability distribution given by:

X	0	1	2	3	4	5	6
P(X)	0.05	0.20	0.25	0.20	0.10	0.15	0.05

Find the probability that between 12:00 noon and 1:00 pm on a particular day, there will be 3 or more calls.

$$P(X \geq 3) = \sum_{X=3}^{6} P(X) = P(3) + P(4) + P(5) + P(6)$$

$$= 0.20 + 0.10 + 0.15 + 0.05$$

$$= 0.5$$

Continuous Random Variables

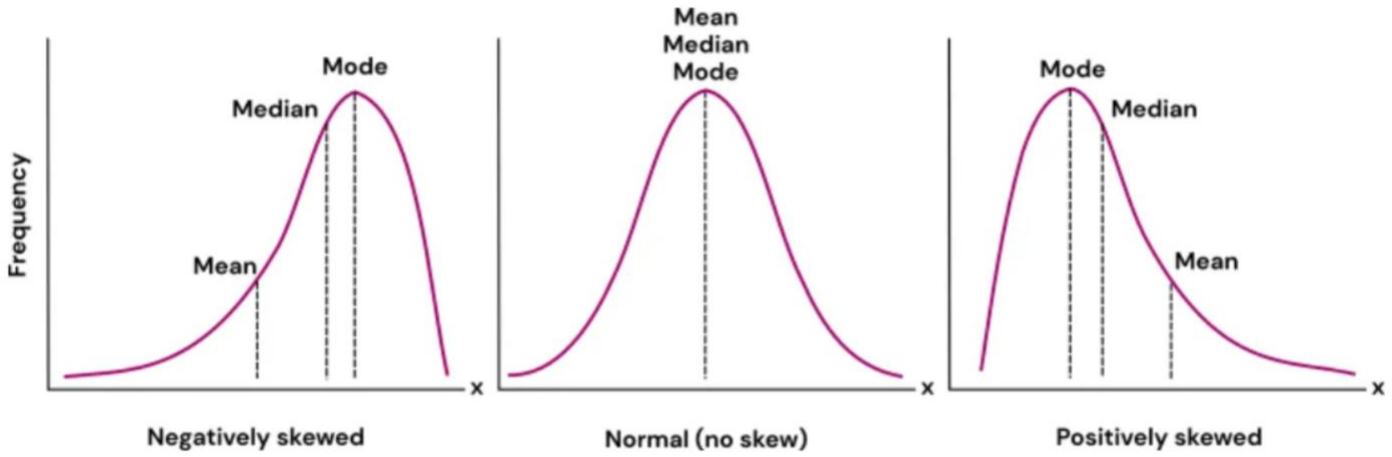
Many real-life business variables approximately follow a normal distribution:

- Waiting time in a bank
- Daily sales revenue
- Share returns
- Amount withdrawn from an ATM
- IQ scores
- Packaging weights
- Employee performance scores
- Standardized examination scores

Probability Distribution
(for Continuous Random Variable)

- The **shape** of a continuous probability distribution may take one of the following:

Left-Skewed Symmetric Right-Skewed



The Normal Distribution

- Bell Shaped
- Symmetrical
- Mean, Median and Mode are equal

Location is determined by the mean, μ

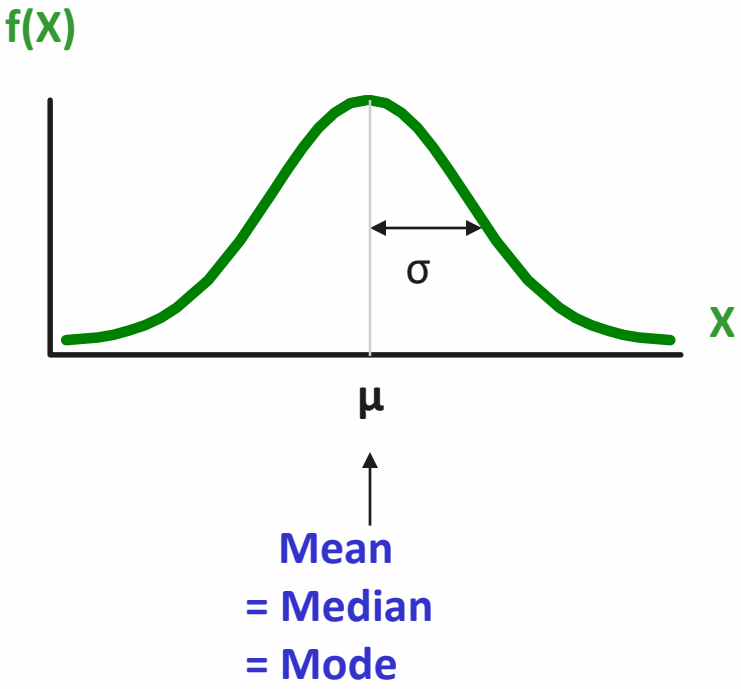
- Changing μ shifts the distribution left or right.

Spread is determined by the standard deviation, σ

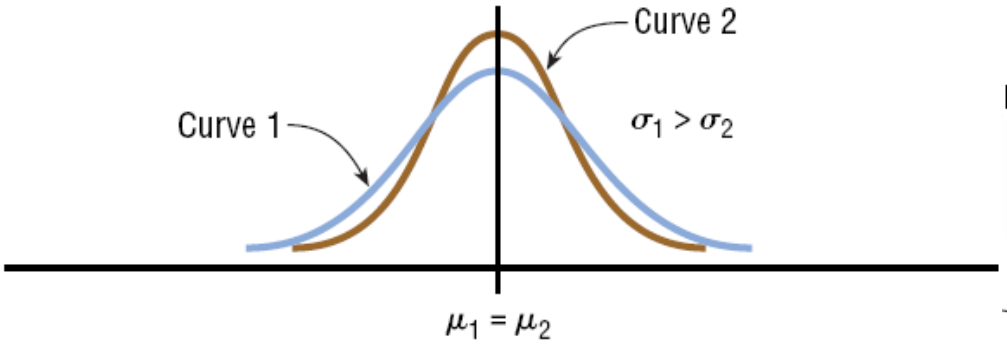
- Changing σ increases or decreases the spread.

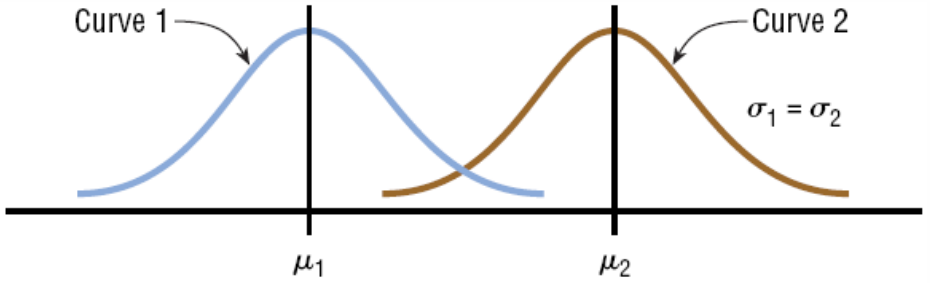
The random variable has an infinite theoretical range:

– ∞ to + ∞

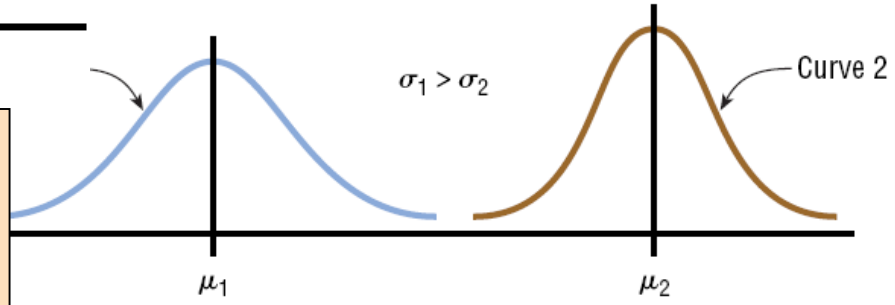


The Normal Distribution





b) Different means but same standard deviations



(c) Different means and different standard deviations

Approximately

68%, 95%, 99.7% of observations

lie within $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$

The Normal Distribution
(Properties)

- The normal distribution curve is bell-shaped.
- The mean, median, and mode are equal and located at the center of the distribution.
- The normal distribution curve is unimodal (i.e., it has only one mode).
- The curve is symmetrical about the mean, which is equivalent to saying that its shape is the same on both sides of a vertical line passing through the center.
- The curve is continuous; that is, there are no gaps or breaks. For every possible value of X, there is a corresponding probability density, f(X).
- The curve never touches the x axis. Theoretically, no matter how far in either direction the curve extends, it never meets the x axis—but it gets increasingly closer.

The Standardized Normal Distribution

- Since each normally distributed variable has its own mean and standard deviation, the shape and location of these curves will vary.
- In practical applications, one would need a table of areas under the curve for each variable.
- To simplify this, statisticians use the **standardized normal distribution**.

Why Standardize?

Bank A customer waiting time:

- Mean = 8 minutes
- Standard deviation = 2 minutes
- Customer at A waited 12 min.

Bank B customer waiting time:

- Mean = 20 minutes
- Standard deviation = 5 minutes
- Customer at B waited 28 min.

Who waited unusually long?

Without standardization, the waiting times cannot be compared directly.

Standardization converts every observation onto a common scale.

The Standardized Normal Distribution

- Any normal distribution (with any mean and standard deviation combination) can be transformed into the **standardized normal distribution (Z)**
- The standardized normal distribution (Z) has a mean of 0 and a standard deviation of 1.
- For example:
  - Mean exam score = 60
    - Student A = 62
    - Student B = 93
  - Obviously, 92 is much farther from the average.
- The standard score (Z-score) measures how many standard deviations an observation is from the mean

Translation to the Standardized Normal Distribution

- Translate from X to the standardized normal (the "Z" distribution) by subtracting the mean of X and dividing by its standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

The Z distribution always has mean = 0 and standard deviation = 1

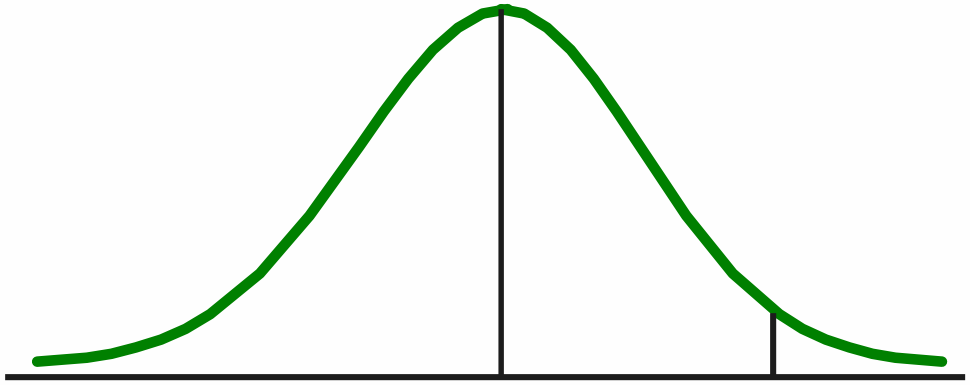
Example Standardized Normal

- If X is distributed normally with mean of ₵100 and standard deviation of ₵50, the Z value for X = ₵200 is

$$Z = \frac{X - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$

- This says that X = ₵200 is two standard deviations (2 increments of ₵50 units) above the mean of ₵100.

Comparing X and Z units



¢100 ¢200 ¢X (μ = ¢100, σ = ¢50)

0 2.0 Z (μ = 0, σ = 1)

Note that the shape of the distribution is the same, only the scale has changed. We can express the problem in the original units (X in Ghana cedis) or in standardized units (Z)

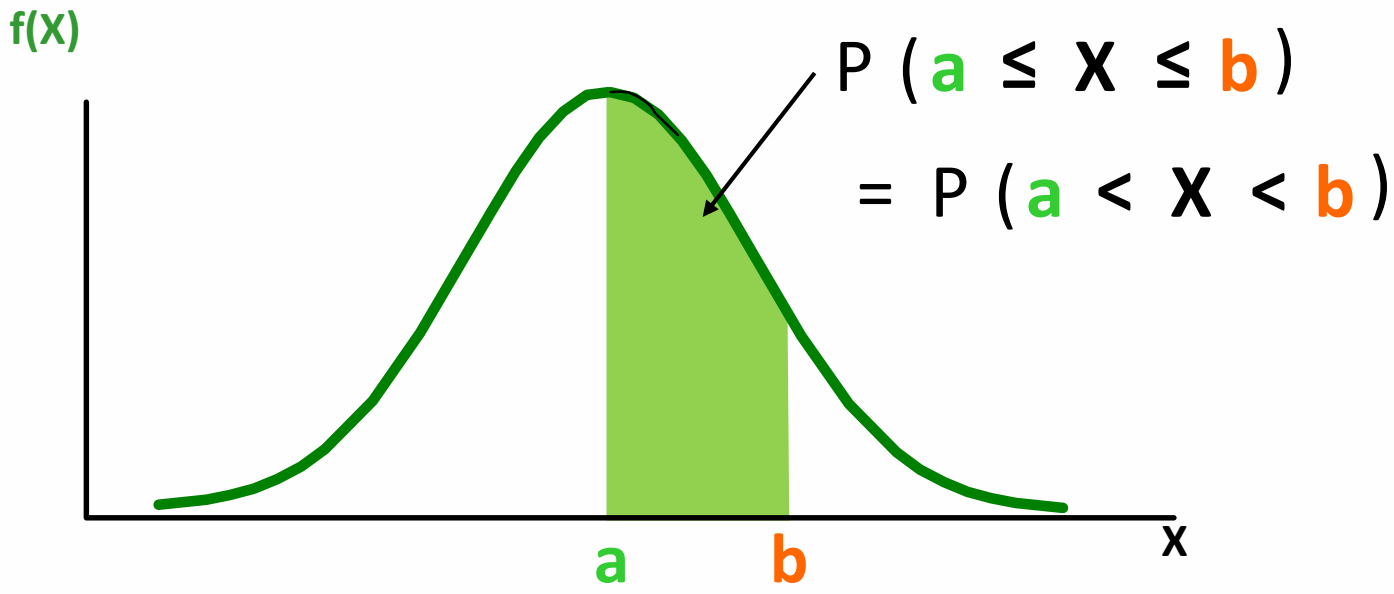
Interpreting Z-scores

- Positive Z → Above the mean
- Negative Z → Below the mean
- Z = 0 → Exactly at the mean

Z-score	Meaning
0	Exactly average
+1	One SD above average
+2	Two SD above average
-1	One SD below average
-2	Two SD below average

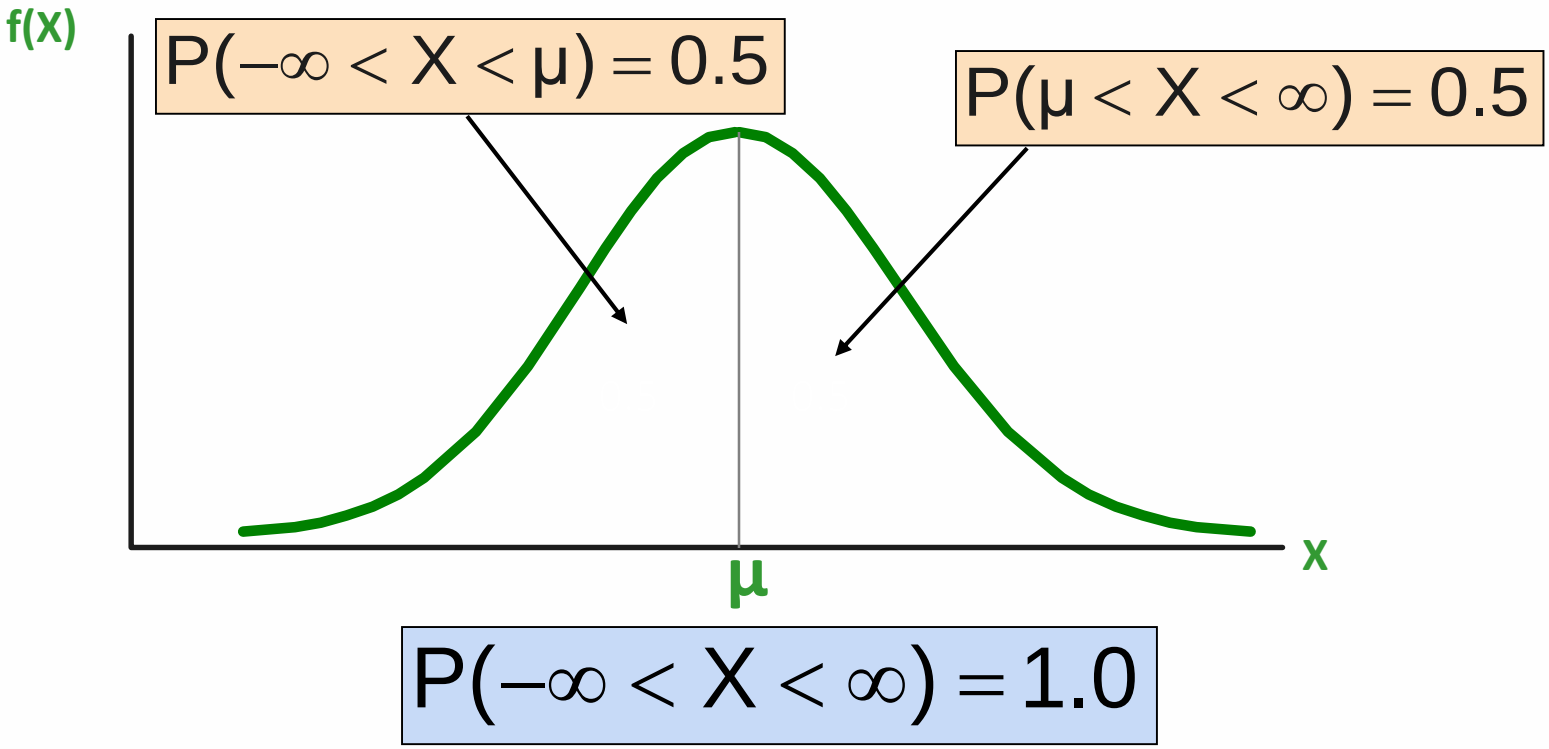
Finding Normal Probabilities

Probability is measured by the area under the curve



Probability as Area Under the Curve

The total area under the curve is 1.0, and the curve is symmetric, so half is above the mean, half is below



The Standardized Normal Table

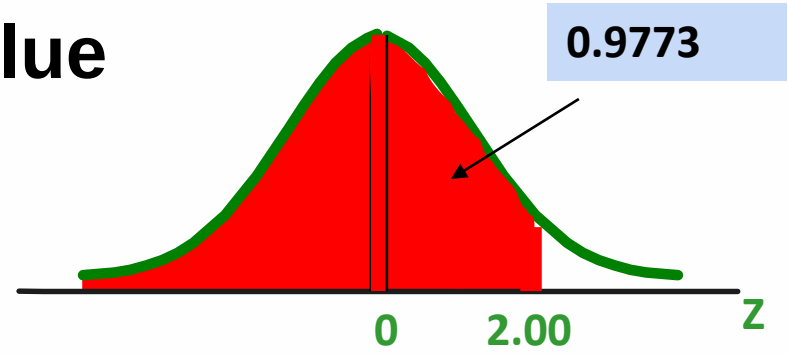
1. To the left of any *z* value

Example:

$P(Z < 2.00) = P(0 < Z < 2.00) + 0.5$

$= 0.4773 + 0.5$

$= 0.9773$

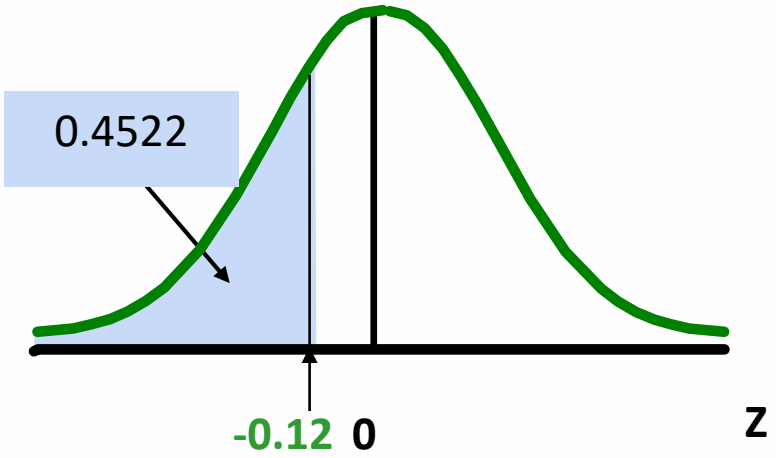


Example:

$P(Z < -0.12) = 0.5 - P(-0.12 < Z < 0)$

$= 0.5 - 0.0478$

$= 0.4522$



The Standardized Normal Table

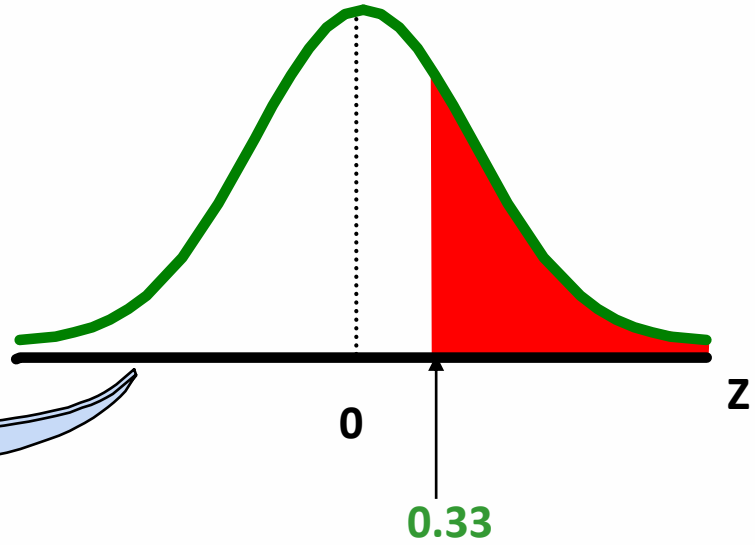
2. To the right of any z value:

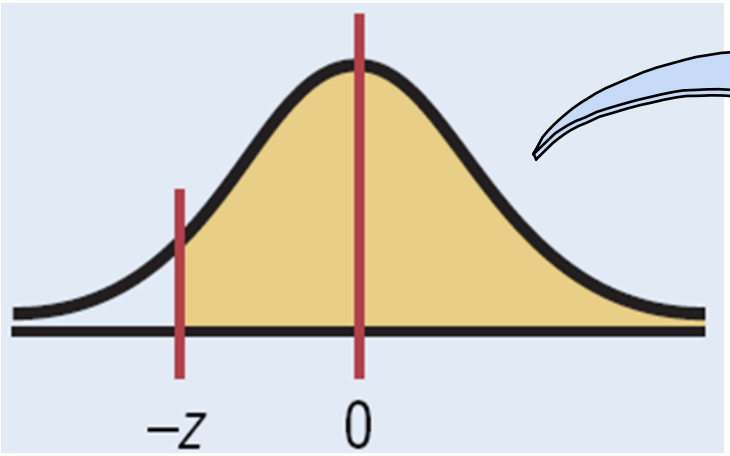
Example:

$P(Z > 0.33) = 0.5 - P(0 < Z < 0.33)$

$= 0.5 - 0.1293$

$= 0.3707$



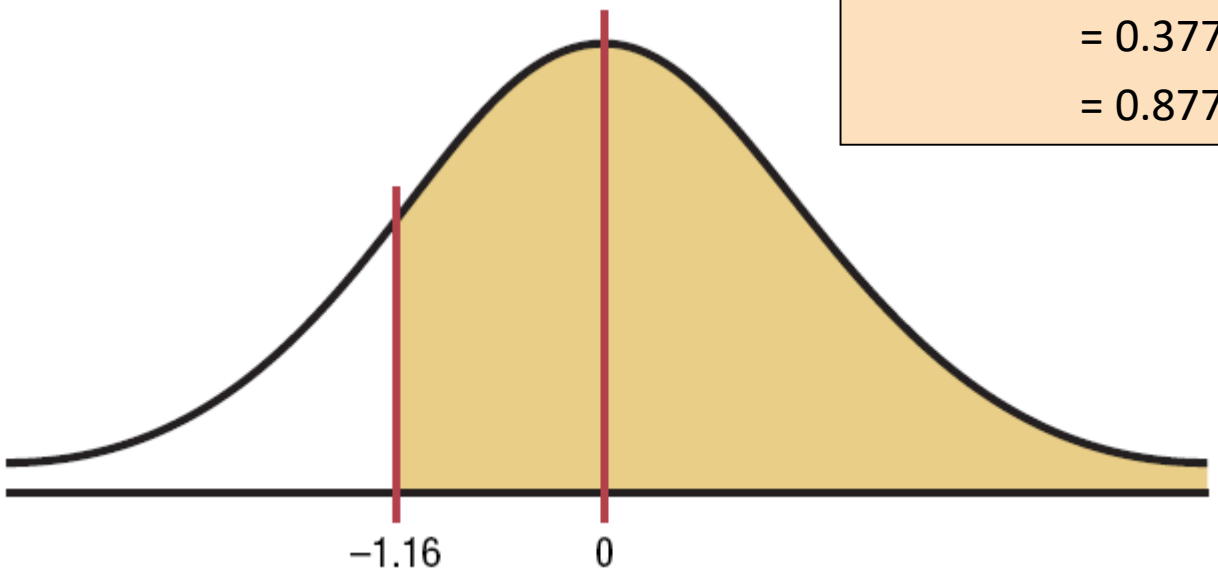


For the figure on the left we add 0.5 to the area between 0 and Z.

The Standardized Normal Table

Example:

Find the area to right of $z = -1.16$.

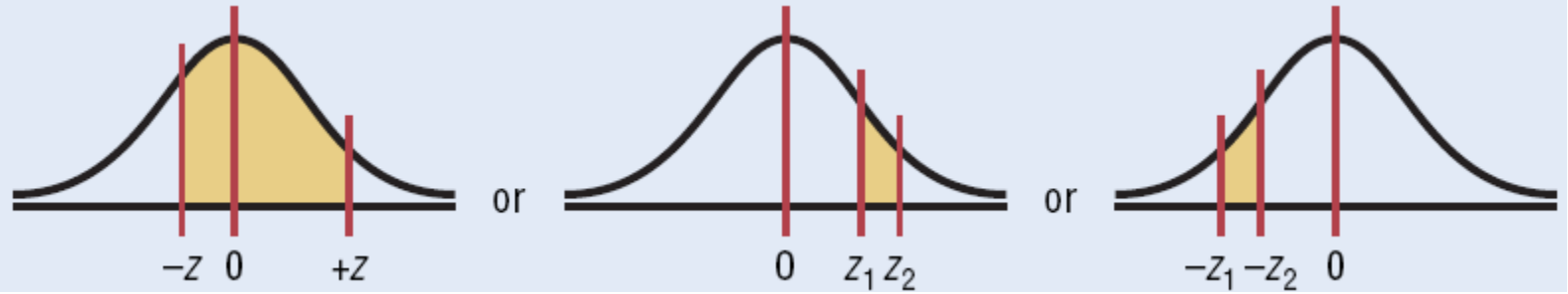


$$P(Z > -1.16) = P(-1.16 < Z < 0) + 0.5$$
$$= 0.3770 + 0.5$$
$$= 0.8770$$

The Standardized Normal Table

3. Between two z values:

- For the figure on the far left, we add the two areas.
- For the other two, we subtract the corresponding areas



Example 7.1

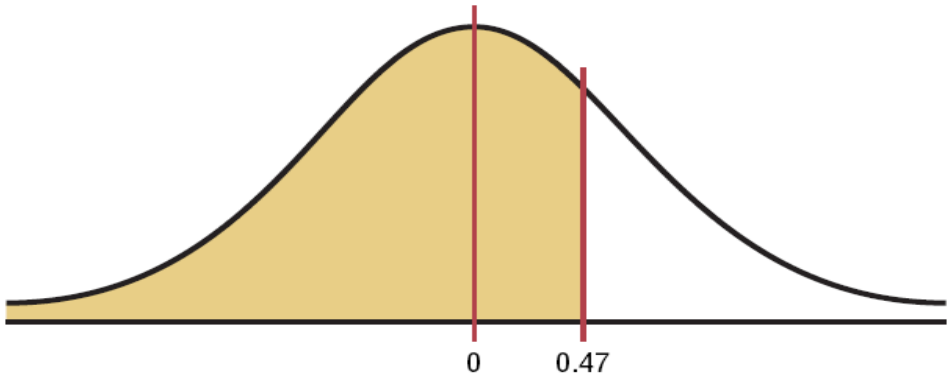
A survey by the Ghana Tourism Authority found that tourists spend, on average, $146.21 per day during the Christmas holidays. Assume the standard deviation is $29.44. Find the percentage of tourists who spend less than $160.00 per day. Assume the variable is normally distributed.

Solution: Example 7.1

Step 1: Find the *z* value corresponding to $160.00.

$$z = \frac{X - \mu}{\sigma} = \frac{160.00 - 146.21}{29.44} = 0.47$$

Step 2: Draw the normal distribution curve.



Step 3: Find the area to the left of *z* = 0.47.

The area between 0 and 0.47 is 0.1808. The total area is 0.6808. i.e. 68% of tourists spend less than $160.

Example 7.2

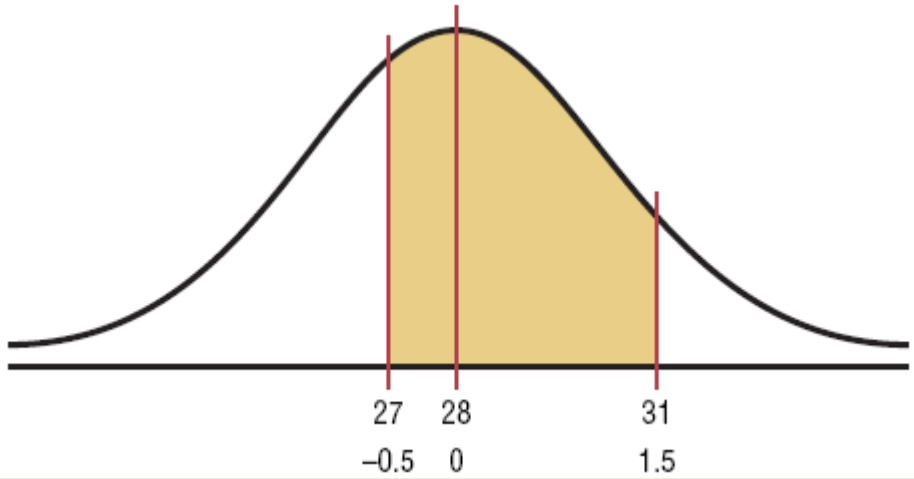
Each month, a Ghanaian household generates an average of 28 pounds of newspaper for garbage or recycling. Assume the standard deviation is 2 pounds. If a household is selected at random, find the probability that it generates between 27 and 31 pounds per month. Assume the variable is approximately normally distributed.

Solution: Example 7.2

- Find *z* values corresponding to 27 and 31.

$$z = \frac{27 - 28}{2} = -0.5 \qquad z = \frac{31 - 28}{2} = 1.5$$

- Find the area between *z* = -0.5 and *z* = 1.5.



The total area is 0.1915 +0.4332 which is 0.6247.

The probability is 0.6247, or approximately 62.47%.

Example 7.3

Students are usually tested on various subjects to determine their level of intelligence. In an intelligence examination administered to 2,000 students of the BBA programmes, the average score recorded by the Lecturer was 42, and the standard deviation of the scores from their mean was 24. Assuming that the distribution of scores is normal, find:

(a) The number of students whose score exceeds 50.
(b) The number of students whose score is between 30 and 54 inclusive.
(c) The score corresponding to the standardized normal value of 2.
(d) What is the probability of a score which is less than 70?
(e) How many students would have a score of at least 66?

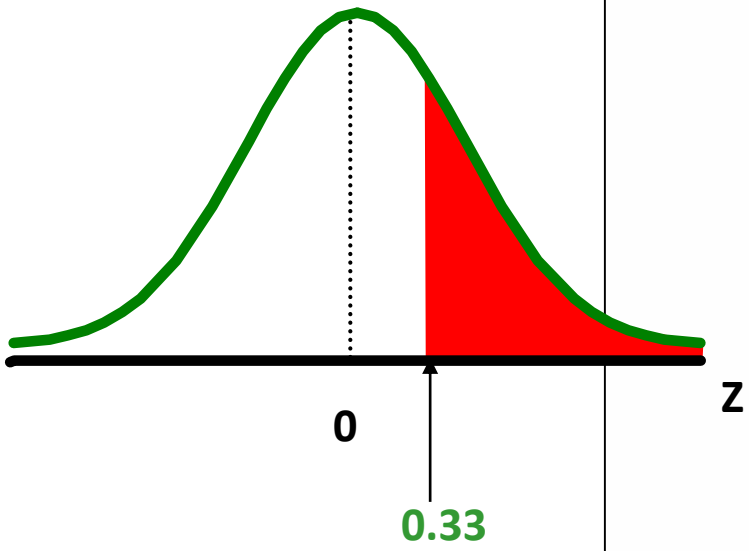
Solution: Example 7.3

Given that $\mu = 42$ and $\sigma = 24$. The standard normal distribution is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 42}{24}.$$

(a) The standard normal distribution for the # of students to exceed the score of 50 is

$$Z = \frac{50 - 42}{24} = 0.33$$



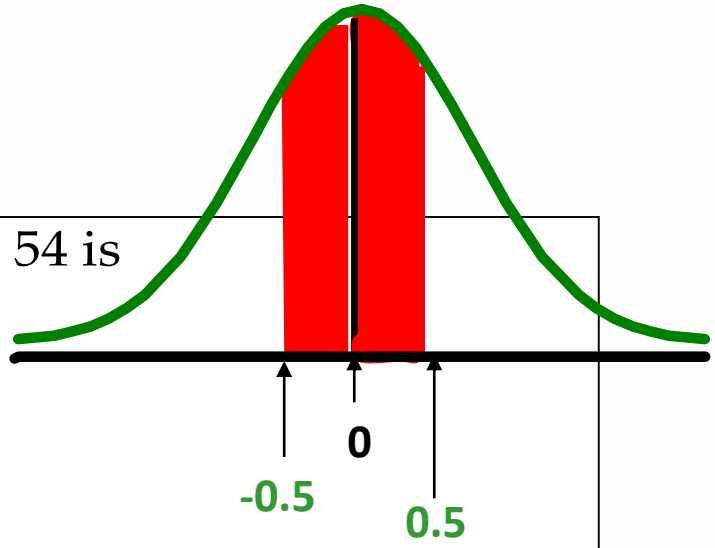
$$P(\text{a student score exceeds } 50) = P(X > 50) = P(Z > 0.33)$$
$$= 1 - P(Z \le 0.33)$$
$$= 1 - 0.6293$$
$$= 0.3707$$

Therefore, out of the 2000 students, the number of students exceeding a score of 50

$= 2000 \times 0.3707$

$= 741.4 \approx 741$

Solution: Example 7.3



(b) The # of students with score b|n 30 and 54 is

$$X = 30 \Rightarrow \quad Z = \frac{30-42}{24} = -0.5$$

$$X = 54 \Rightarrow \quad Z = \frac{54-42}{24} = 0.5$$

$$P(\text{a student scores b|n 30 and 54}) = P(30 \le X \le 54) = P(-0.5 \le Z \le 0.5)$$

$$= P(Z < 0.5) - P(Z < -0.5)$$

$$= 0.6915 - 0.3085$$

$$= 0.383$$

Therefore, out of the 2000 students, the number of students whose score is b|n 30 and 54

$$= 2000 \times 0.383$$

$$= 766$$

Solution: Example 7.3

(c) The score corresponding to the standardized normal score of 2 is given by

$$2 = \frac{X - 42}{24}$$

$$2 \times 24 = X - 42$$

$$X = 48 + 42$$

$$X = 90$$

The students with a standardised score of 2 will have a score of 90.

Solution: Example 7.3

(d) The probability of a student getting a score of less than 70 is given by

$$X = 70 \Rightarrow Z = \frac{70 - 42}{24} = 1.17$$

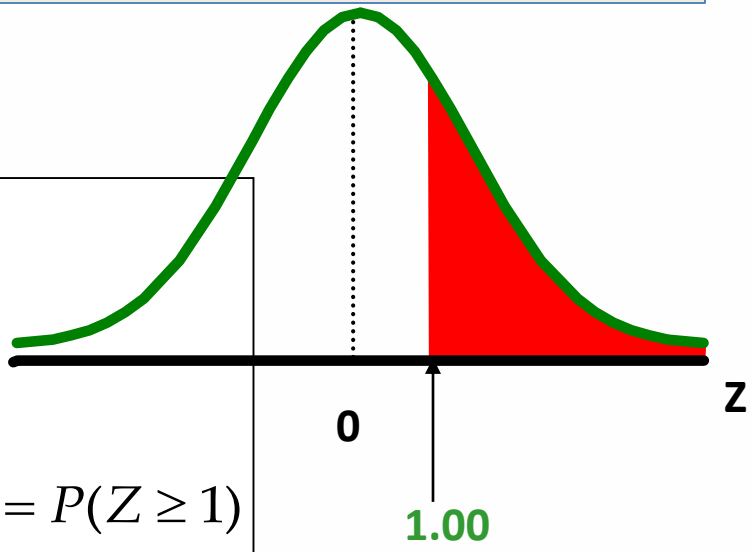
$$\begin{aligned} \text{P(a student score less than 70)} &= P(X < 70) \\ &= P(Z < 1.17) \\ &= 0.8790 \end{aligned}$$

Solution: Example 7.3

(e) The standard normal distribution for the # of students scoring at least 66,

$$X = 66 \Rightarrow Z = \frac{66 - 42}{24} = 1$$

$$P(\text{a student score at least 66}) = P(X \geq 66) = P(Z \geq 1)$$
$$= 1 - P(Z < 1)$$
$$= 1 - 0.8413$$
$$= 0.1587$$



Therefore, out of the 2000 students, the number of students who score at least 66

$= 2000 \times 0.1587$

$= 317.4 \approx 317.$

Exercise 7.1

The Ghana Fire Service reports that the average time it takes to respond to an emergency call is 25 minutes. Assume the variable is approximately normally distributed, with a standard deviation of 4.5 minutes. If 80 calls are randomly selected, approximately how many will be responded to in less than 15 minutes?

Exercise 7.2

The total cholesterol values for a certain population are approximately normally distributed with a mean of 200 mg/100 ml and a standard deviation of 20 mg/100 ml. If 1,500 people were drawn from this population, find how many have cholesterol value

(a) between 180 and 200 mg/100 ml,

(b) greater than 225 mg/100 ml,

(c) less than 150 mg/100 ml,

(d) between 160 and 210 mg/100 ml.

Exercise 7.3

The Marketing Manager of Marina Mall records weekly purchases of customers. Last week, the total purchases were GH¢ 2,000. The customer accounts records at the Mall have an average purchase of GH¢48 and a variance of GH¢256. Assuming that the purchases are normally distributed:

(a) What amount of the total purchases is over a customer's purchase of GH¢60?

(b) What amount of the total purchases is between GH¢50 and GH¢80 purchases inclusive per customer?

(c) What amount of the purchases is less than GH¢10?

THANK YOU!